DESCENT OF A FAMILY OF RELATIVE CYCLES TO A NOETHERIAN MODEL, CRITERION VIA COMPLETELY DECOMPOSED PROPER MORPHISMS

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Allowed standard prerequisites (editorial).

- Existence of a completely decomposed proper morphism g_0 : S'_0 to S_0 putting a given family on a Noetherian base into the image of the cycle-classes construction (relative-cycles chapter, Lemma lemma-get-cycles, canonical tag 0H59). cited at proof lines 3179-3181. Direction (1) to (2): produces the witness for the Noetherian model, which base changes to S.
- The cycle-classes construction whose image is being characterized (relative-cycles chapter, Equation equation-cycle-classes-general, canonical tag 0H6V). cited at proof lines 3173, 3181 and 3188. The map z(X/S, r) is characterized against; being in its image is condition (2).
- Approximation: a quasi-compact quasi-separated scheme is a directed limit of finite type schemes over Z with affine transitions (limits chapter, Proposition limits-proposition-approximate, canonical tag 01ZA). cited at proof line 3197. Sets up writing S = lim S_i for the descent of the data in direction (2) to (1).
- Descent of finite presentation along limits (limits chapter, Lemma limits-lemma-descend-finite-presentation, canonical tag 01ZM). cited at proof line 3208. One of the six Limits inputs descending the morphism and subscheme data to some finite stage.
- Descent of closed immersions of finite presentation along limits (limits chapter, Lemma limits-lemma-descend-closed-immersion-finite-presentation, canonical tag 01ZP). cited at proof line 3209. Descends the closed subschemes Z_a to Z_{ai}.
- Descent of flatness of finite presentation along limits (limits chapter, Lemma limits-lemma-descend-flat-finite-presentation, canonical tag 04AI). cited at proof line 3210. Preserves the flatness of Z_{ai} over S'_i.
- Descent of surjectivity along limits (limits chapter, Lemma limits-lemma-descend-surjective, canonical tag 07RR). cited at proof line 3211. Preserves complete decomposition of the descended g_i.

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- Eventual properness along limits (limits chapter, Lemma limits-lemma-eventually-proper, canonical tag 081F). cited at proof line 3212. Makes the descended g_i proper at some finite stage.
- Dimension bounds along limits (limits chapter, Lemma limits-lemma-limit-dimension, canonical tag 05M5). cited at proof line 3213. Preserves the relative dimension bound of the descended subschemes.
- Descent of completely decomposed morphisms (more-morphisms chapter, Lemma more-morphisms-lemma-descend-cd, canonical tag 0GTM). cited at proof line 3216. Makes the descended g_i completely decomposed.
- Closedness of the locus where two relative cycles on the fibres agree (relative-cycles chapter, Lemma lemma-relative-cycles-equal, canonical tag 0H5C). cited at proof line 3232. The equal-locus E is closed, enabling the constructible-limit argument.
- A morphism from a limit scheme landing in a constructible subset factors through some finite stage (limits chapter, Lemma limits-lemma-limit-contained-in-constructible, canonical tag 05F4). cited at proof lines 3234-3236. Promotes the inclusion of S" into E to a finite stage i', making the two pullbacks literally equal.
- Unique descent of a family of cycles on fibres along a completely decomposed proper morphism (relative-cycles chapter, Lemma lemma-descend-family, canonical tag 0H4X). cited at proof lines 3239-3242. Produces the unique family alpha_i on X_i/S_i with g_i^* alpha_i = alpha'_i.
- The h-descent property placing descended families into the relative cycles group z(X_i/S_i, r) (relative-cycles chapter, Lemma lemma-relative-cycles-h-descent, canonical tag 0H52). cited at proof lines 3244-3245. Verifies alpha_i is an honest relative r-cycle, completing condition (1).

Source reference. Original lemma 0H59: sources/stacks-chow/tags.txt, source lines 21379–21379. Cited at proof lines 3179-3181: direction (1) to (2) rests on it.

Source reference. Original equation 0H6V: sources/stacks-chow/tags.txt, source lines 21435–21435. Cited at proof lines 3173, 3181, 3188: the image condition (2) is formulated against it.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cited at proof line 3197: writes S as a limit of Noetherian finite type stages.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof line 3208: descends the morphism data to a finite stage.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cited at proof line 3209: descends the closed subschemes Z_a.

Source reference. Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cited at proof line 3210: keeps the descended subschemes flat.

Source reference. Original lemma 07RR: sources/stacks-chow/tags.txt, source lines 9876–9876. Cited at proof line 3211: keeps the descended cover surjective.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof line 3212: makes the descended g_i proper.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Cited at proof line 3213: keeps the relative dimension bound of the descended subschemes.

Source reference. Original lemma 0GTM: sources/stacks-chow/tags.txt, source lines 20972–20972. Cited at proof line 3216: makes the descended g_i completely decomposed.

Source reference. Original lemma 0H5C: sources/stacks-chow/tags.txt, source lines 21382–21382. Cited at proof line 3232: makes the equal locus E closed.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cited at proof lines 3234-3236: promotes the equal-locus inclusion to a finite stage.

Source reference. Original lemma 0H4X: sources/stacks-chow/tags.txt, source lines 21367–21367. Cited at proof lines 3239-3242: descends alpha'_i uniquely to alpha_i.

Source reference. Original lemma 0H52: sources/stacks-chow/tags.txt, source lines 21372–21372. Cited at proof lines 3244-3245: verifies alpha_i lies in z(X_i/S_i, r).

Lemma 0.2. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to S$ be a morphism of finite presentation. Let $r \geq 0$ and let α be a family of r-cycles on fibres of X/S. The following are equivalent*

(1) *there exists a cartesian diagram*

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_0 \end{array}$$

where $X_0 \to S_0$ is a finite type morphism of Noetherian schemes and $\alpha_0 \in z(X_0/S_0, r)$ such that α is the base change of α_0 by $S \to S_0$

(2) *there exists a completely decomposed proper morphism $g : S' \to S$ of finite presentation such that $g^*\alpha$ is in the image of (0H6V).*

Proof. Let a diagram and $\alpha_0 \in z(X_0/S_0, r)$ as in (1) be given. By Lemma 0H59 there exists a completely decomposed proper morphism $g_0 : S_0' \to S_0$ such that $g_0^*\alpha_0$ is in the image of (0H6V). Namely, since S_0' is Noetherian, every closed subscheme of $S_0' \times_{S_0} X_0$ is of finite presentation over S_0'. Setting $S' = S \times_{S_0} S_0'$ and using base change by $S' \to S_0'$ we see that (2) holds.

Conversely, assume that (2) holds. Choose a completely decomposed proper morphism $g : S' \to S$ of finite presentation such that $g^*\alpha$ is in the image of (0H6V). Set $X' = S' \times_S X$. Write $g^*\alpha = \sum n_a[Z_a/X'/S']_r$ for some $Z_a \subset X'$ closed subscheme flat, of finite presentation, and of relative dimension $\leq r$ over S'.

Write $S = \lim S_i$ as a directed limit with affine transition morphisms with S_i of finite type over $\mathbf{Z}$, see Limits, Proposition 01ZA. We can find an i large enough such that there exist

(1) a completely decomposed proper morphism $g_i : S_i' \to S_i$ whose base change to S is $g : S' \to S$,

(2) setting $X'_i = S'_i \times_{S_i} X_i$ closed subschemes $Z_{ai} \subset X'_i$ flat and of relative dimension $\leq r$ over S'_i whose base change to S' is Z_a.

To do this one uses Limits, Lemmas 01ZM, 01ZP, 04AI, 07RR, 081F, and 05M5 and and More on Morphisms, Lemma 0GTM. Consider $\alpha'_i = \sum n_a[Z_{ai}/X'_i/S_i]_r \in z(X'_i/S'_i, r)$. The base change of α'_i to a family of r-cycles on fibres of X'/S' agrees with the base change $g^*\alpha$ by construction.

Set $S''_i = S'_i \times_{S_i} S'_i$ and $X''_i = S''_i \times_{S_i} X_i$ and set $S'' = S' \times_S S'$ and $X'' = S'' \times_S X$. We denote $\mathrm{pr}_1, \mathrm{pr}_2 : S'' \to S'$ and $\mathrm{pr}_1, \mathrm{pr}_2 : S''_i \to S'_i$ the projections. The relative r-cycles $\mathrm{pr}_1^*\alpha'_i$ and $\mathrm{pr}_1^*\alpha'_i$ on X''_i/S''_i base change to the same family of r-cycles on fibres of X''/S'' because $\mathrm{pr}_1^*g^*\alpha = \mathrm{pr}_1^*g^*\alpha$. Hence the morphism $S'' \to S''_i$ maps into $E = \{s \in S''_i : (\mathrm{pr}_1^*\alpha'_i)_s = (\mathrm{pr}_1^*\alpha'_i)_s\}$. By Lemma 0H5C this is a closed subset. Since $S'' = \lim_{i' \geq i} S''_{i'}$ we see from Limits, Lemma 05F4 that for some $i' \geq i$ the morphism $S''_{i'} \to S''_i$ maps into E. Therefore, after replacing i by i', we may assume that $\mathrm{pr}_1^*\alpha'_i = \mathrm{pr}_1^*\alpha'_i$. By Lemma 0H4X we obtain a unique family α_i of r-cycles on fibres of X_i/S_i with $g_i^*\alpha_i = \alpha'_i$ (this uses that $S'_i \to S_i$ is completely decomposed). By Lemma 0H52 we see that $\alpha_i \in z(X_i/S_i, r)$. The uniqueness in Lemma 0H4X implies that the base change of α_i is α and we see (1) holds. □